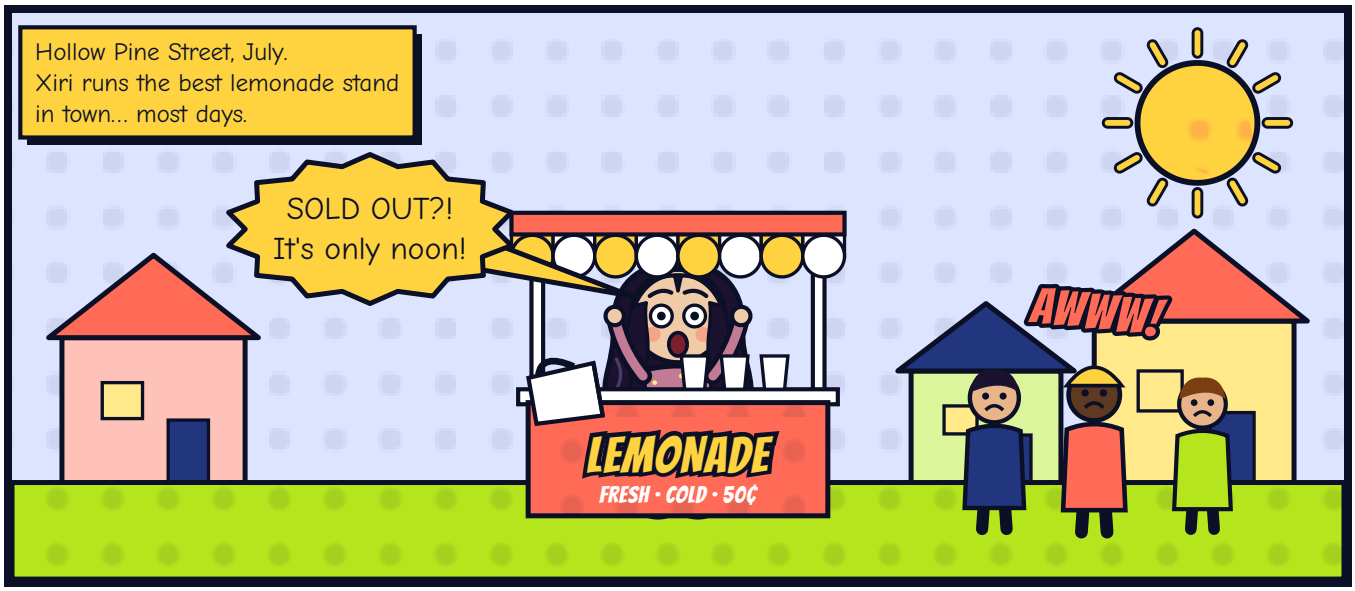
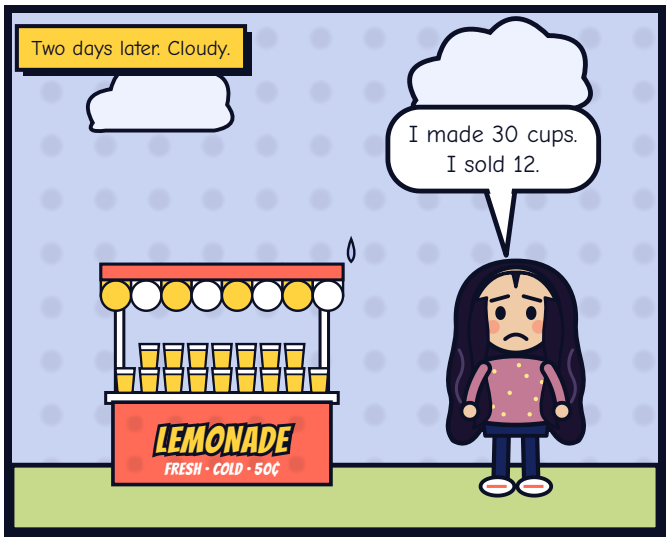
1 · LEMONADE PROBLEM
2 · TWO KNOBS
3 · HOW WRONG
LINE LAB
4 · ERROR HILL
5 · BEST LINE
6 · WHERE IT LIES
CHEAT SHEET
QUIZ

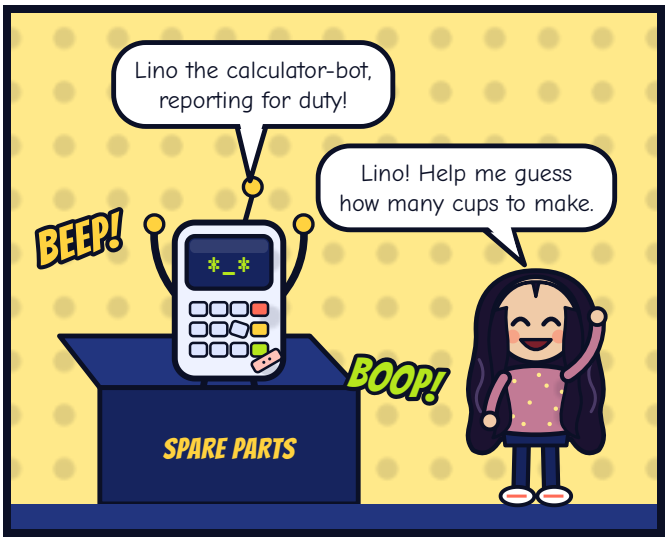
CHAPTER 1

THE LEMONADE PROBLEM

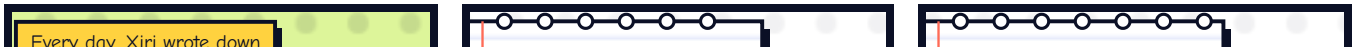
In which Xiri keeps guessing wrong, and a robot shows up.


Hollow Pine Street, July.
Xiri runs the best lemonade stand in town... most days.
SOLD OUT?! It's only noon!
LEMONADE
FRESH · COLD · 50¢
AWWW!


Two days later. Cloudy.
I made 30 cups. I sold 12.
LEMONADE
FRESH · COLD · 50¢


Lino the calculator-bot, reporting for duty!
Lino! Help me guess how many cups to make.
BEEP!
_
BOOP!
SPARE PARTS

SUMMER FAIR!
THIS SATURDAY
Food stalls · Games · Music
Lemonade sellers wanted!
Too few cups: I lose customers. Too many: I waste money.
So we need to... see the FUTURE?
?_?


Every day, Xiri wrote down

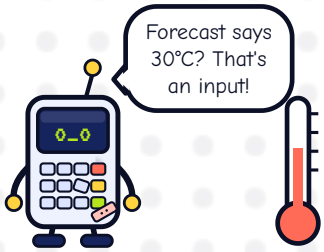
two numbers.

XIRI'S LEMONADE LOG

Temp (°C)	Cups sold
20	12
24	17
28	19
32	25
36	27

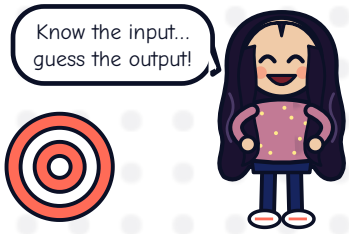
LINO'S NOTEBOOK

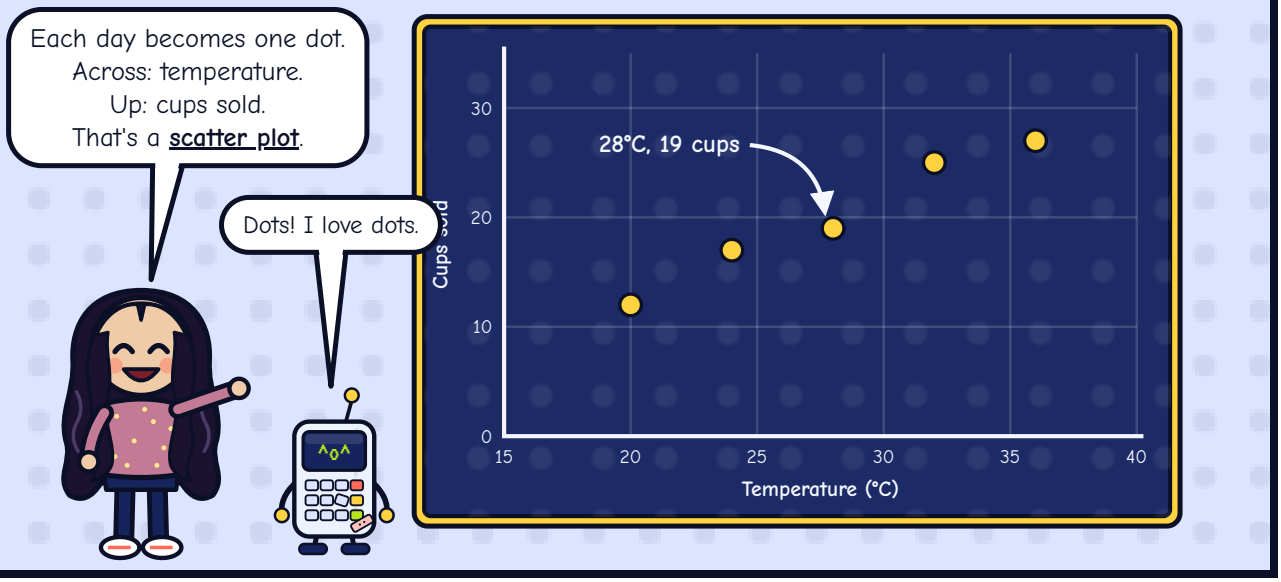
The thing we already KNOW is the **input**.
Here: the temperature.
Also called a **feature**.

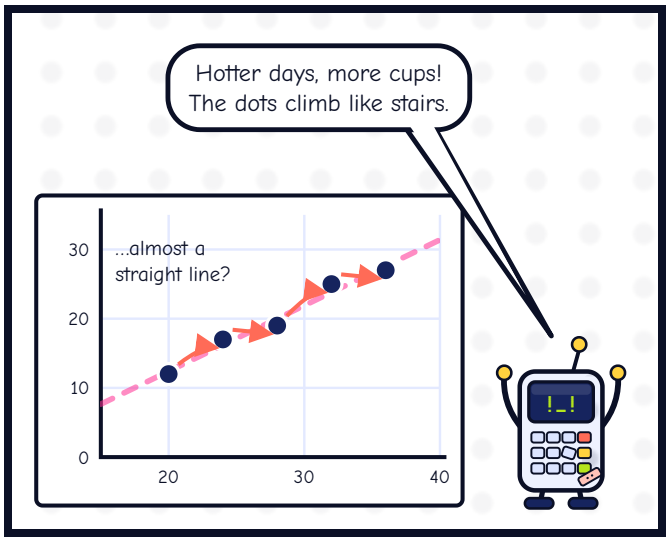


LINO'S NOTEBOOK

The thing we WANT to guess is the **output**.
Here: cups sold.
Also called the **target**.







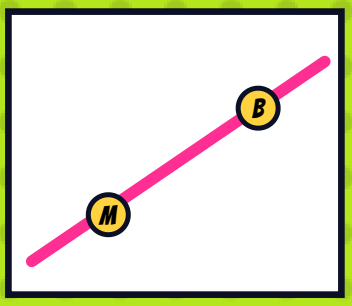
WHAT WE LEARNED

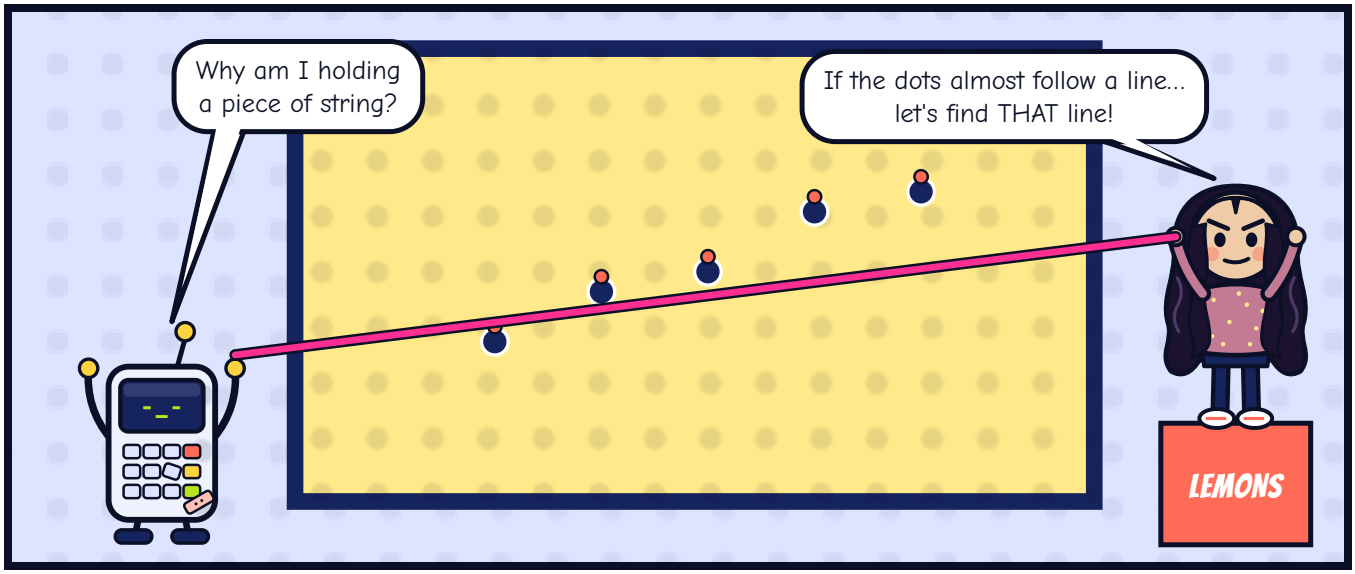
- Data = notes about real days.
- **Input** (feature): temperature.
- **Output** (target): cups sold.
- Hotter days → more cups, almost in a straight line.

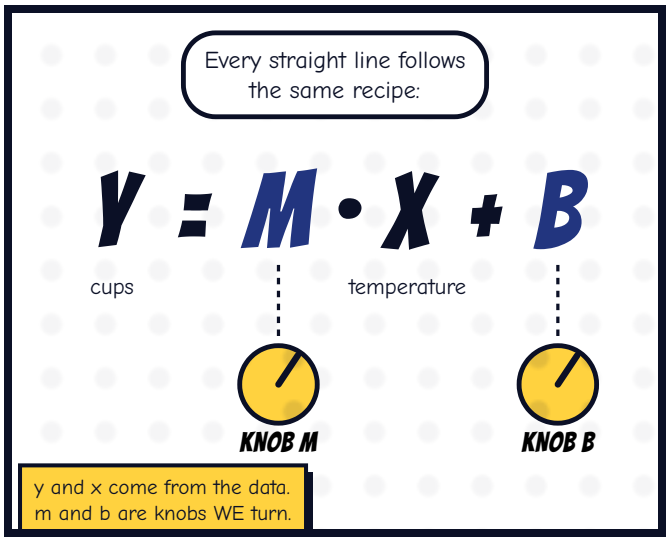
CHAPTER 2

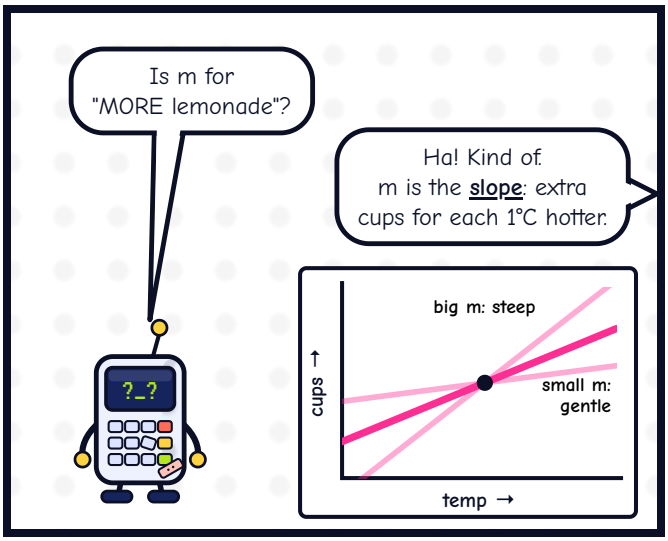
A LINE HAS TWO KNOBS

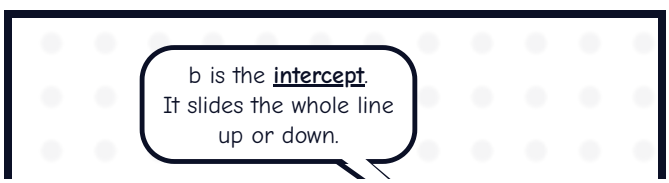
In which a piece of string becomes a math recipe.

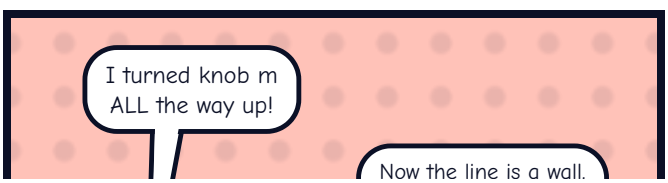


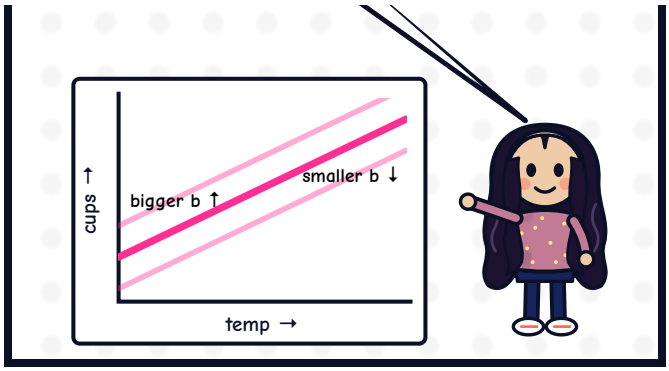


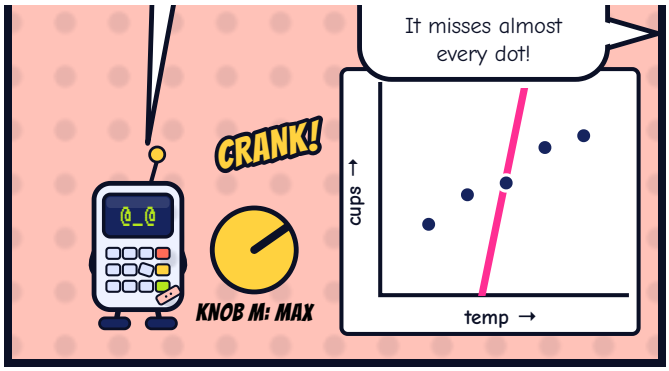


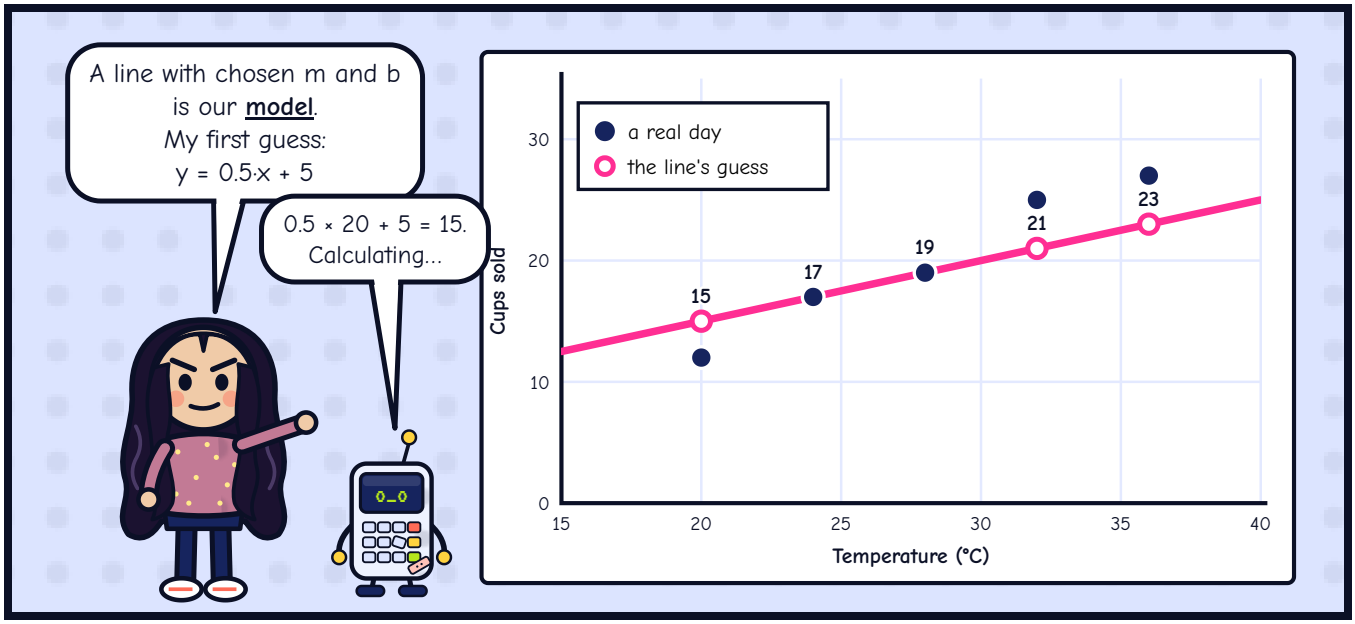












Check the guess, day by day:

Temp	Real cups	Line's guess	Error
20°	12	15	−3
24°	17	17	0
28°	19	19	0
32°	25	21	+4
36°	27	23	+4

Error = real − guess.
20°C: 12 − 15 = −3. Off by 3!

WHAT WE LEARNED

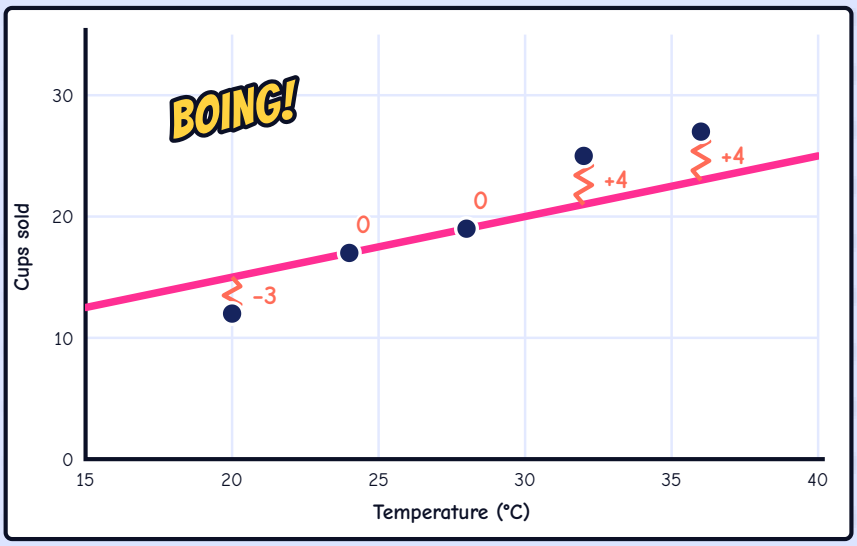
- A line: $y = m \cdot x + b$
- m (slope) tilts it.
- b (intercept) slides it.
- Model = a line with chosen m and b.
- Error = real − guess.

CHAPTER 3

HOW WRONG IS WRONG?

9 16 4

In which mistakes get squared, and Lino learns to count.



Each gap between a dot and the line is a **residual**: the error left over.

Boing!

Easy! Add them up: -3 + 0 + 0 + 4 + 4 = 5

Careful! -10 and +10 add up to 0... but both are big mistakes!

= 0 ?!

-10 +10

Fix: **square** each error. Negatives turn positive, and big misses become BIG squares.

Ouch!

$1^2 = 1$ $(-3)^2 = 9$ $4^2 = 16$

LINO'S NOTEBOOK: MEAN SQUARED ERROR

ERROR = the gaps (residuals): -3, 0, 0, +4, +4

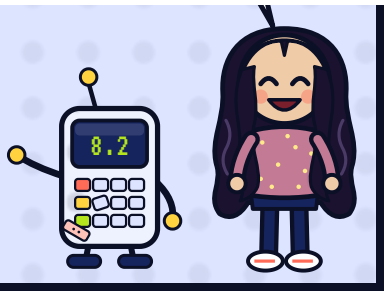
SQUARED = each gap times itself: 9, 0, 0, 16, 16

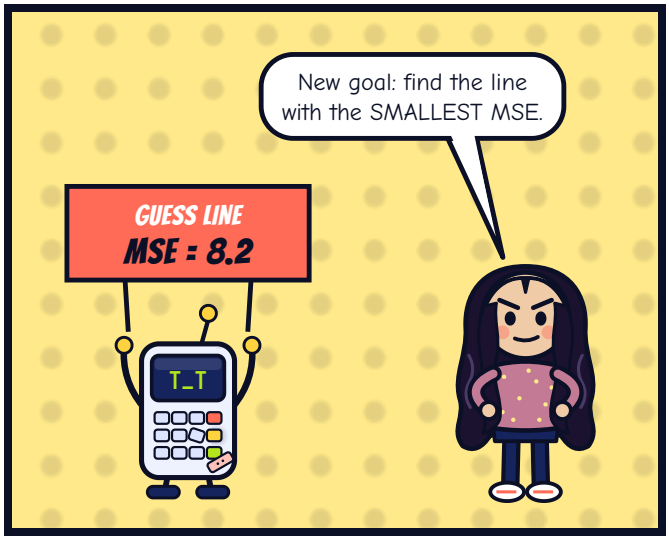
One number for how wrong the line is. Lower = better!

MEAN = the average: add them up, divide by 5 days

$$(9 + 0 + 0 + 16 + 16) \div 5$$

$$= 41 \div 5 = 8.2$$





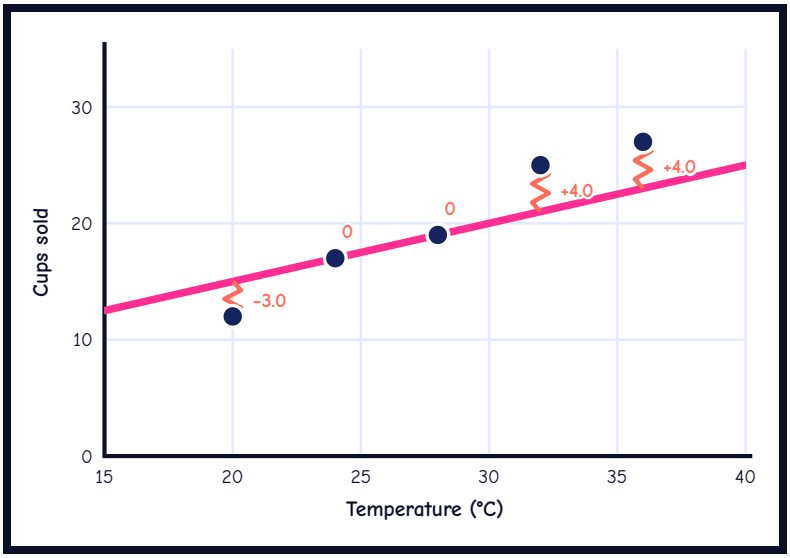
WHAT WE LEARNED

- **Residual** = gap from a dot to the line.
- Square each one: no canceling, big misses count more.
- **MSE** = average of the squared errors.
- Goal: the line with lowest MSE.

INTERMISSION · TRY IT YOURSELF

LINO'S LINE LAB

Turn the two knobs. The pink line moves, the coral springs show each residual, and the MSE updates. Xiri's first guess scored 8.2. Can you get below 1?



KNOB M (SLOPE) 0.50

KNOB B (INTERCEPT) 5.0

MSE 8.20

Xiri's first guess. Plenty of room to improve.

improve.

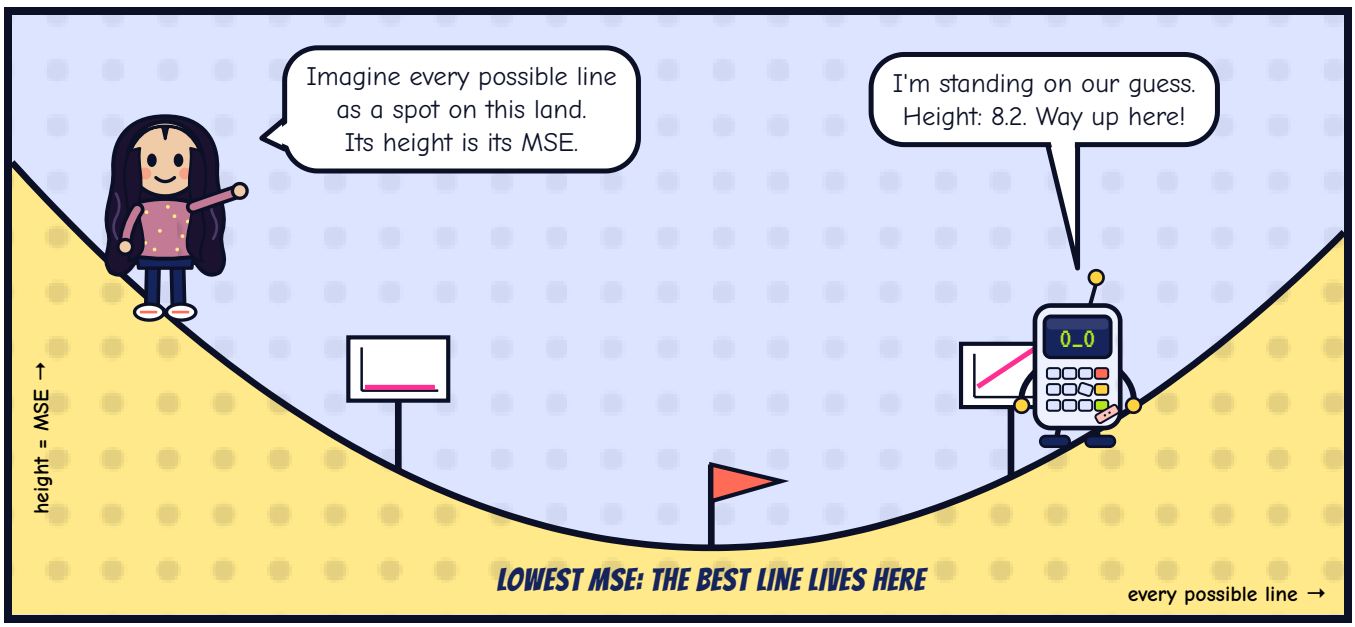
XIRI'S GUESS

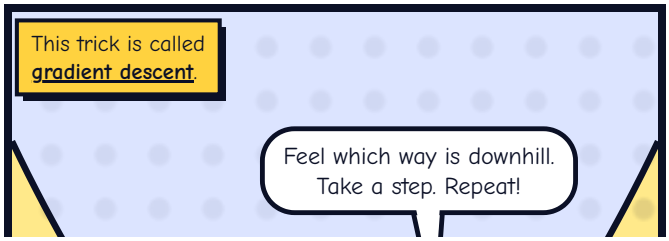
SHOW THE BEST LINE

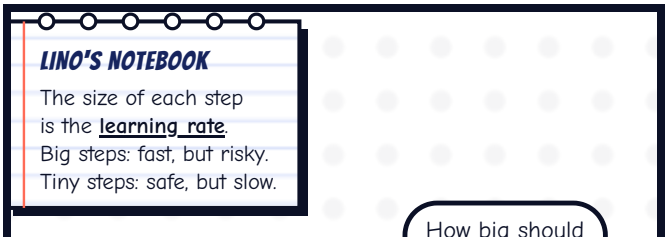
CHAPTER 4

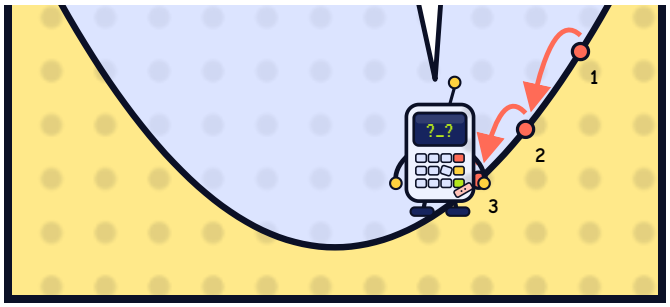
ROLLING DOWN ERROR HILL

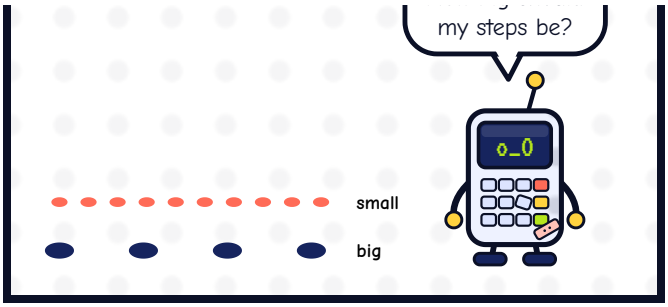
In which Lino walks downhill until he can't anymore.

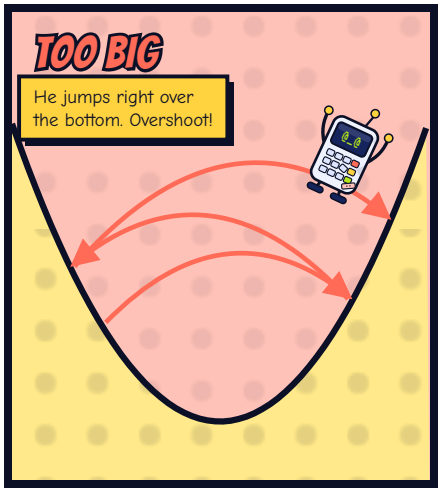


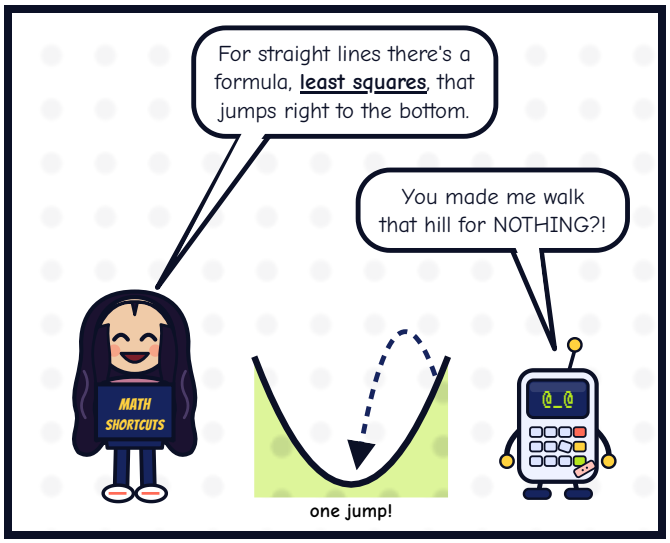


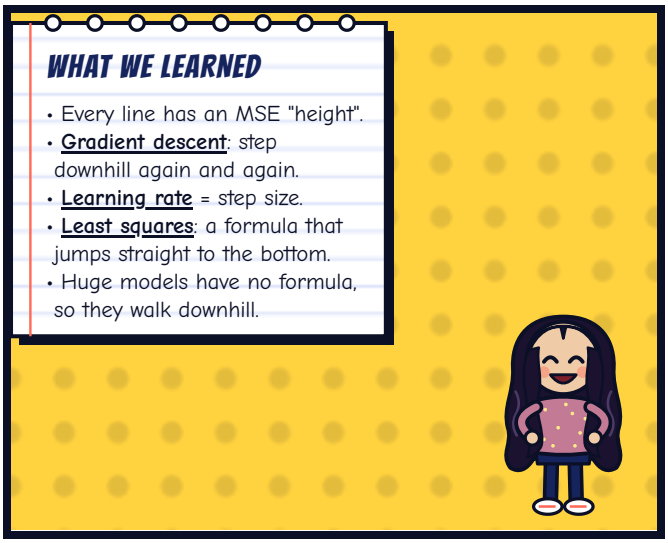






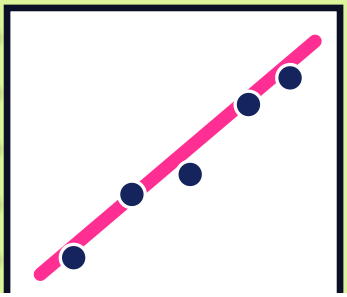




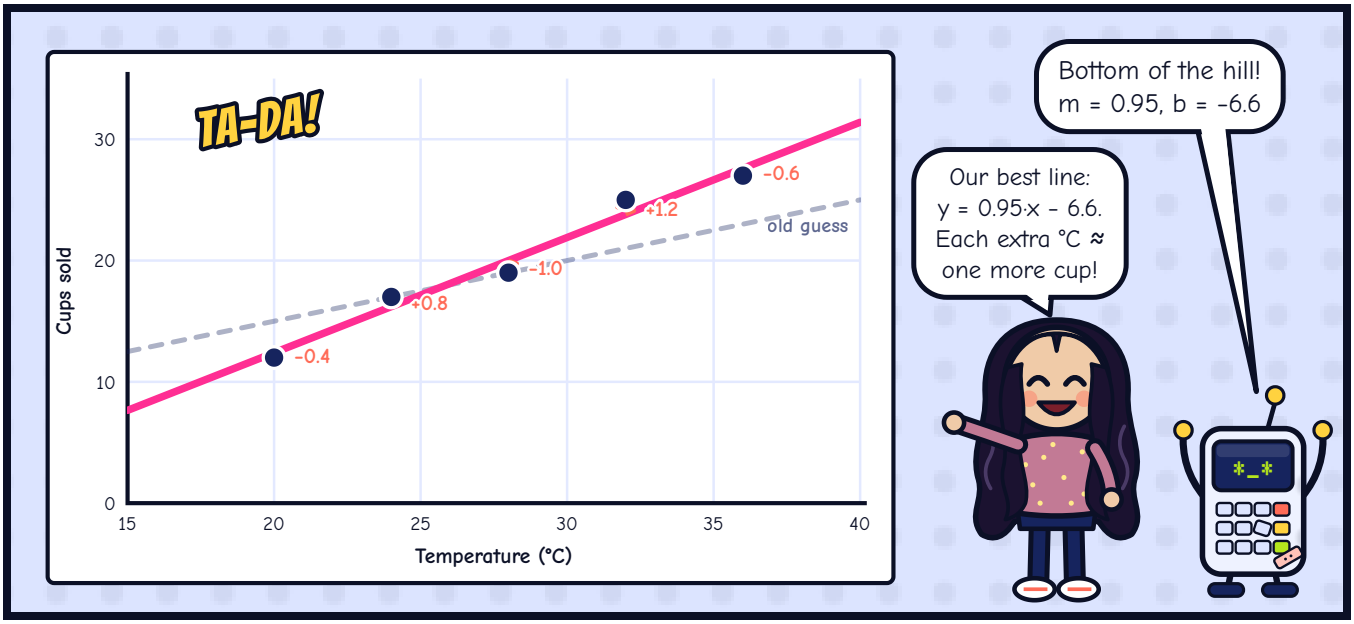


CHAPTER 5

THE BEST LINE



In which the line meets the Summer Fair.

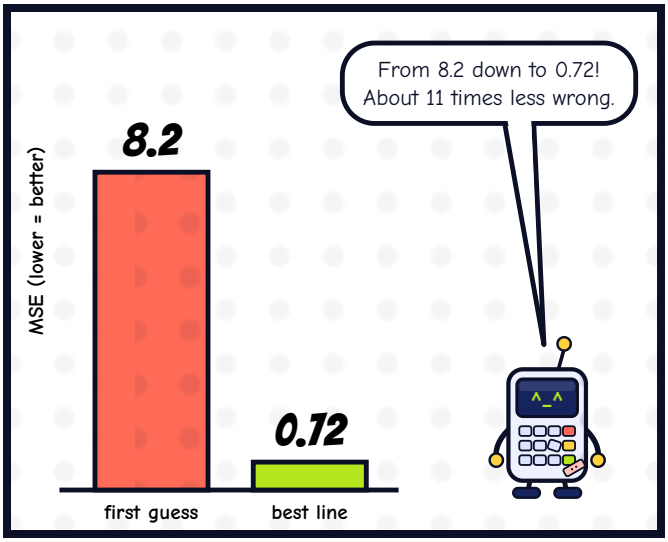


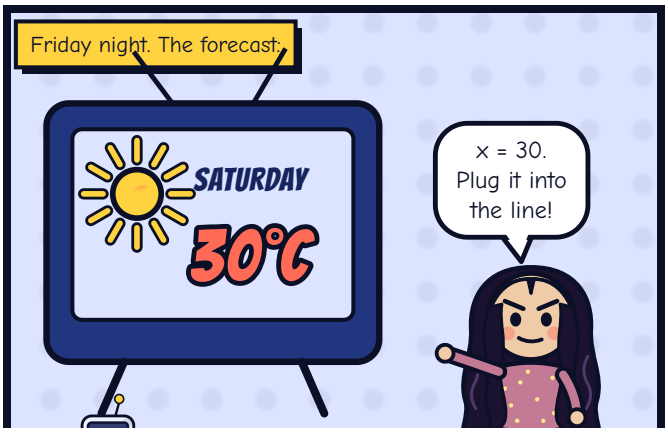
The best line, day by day:

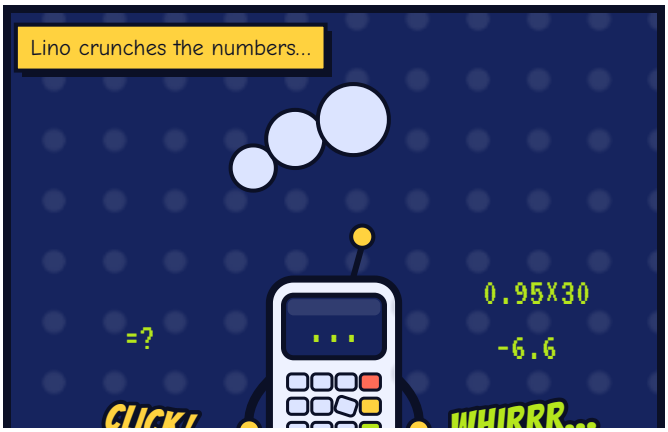
Temp	Real	Line says	Residual	Squared
20°	12	12.4	−0.4	0.16
24°	17	16.2	+0.8	0.64
28°	19	20.0	−1.0	1.00
32°	25	23.8	+1.2	1.44
36°	27	27.6	−0.6	0.36

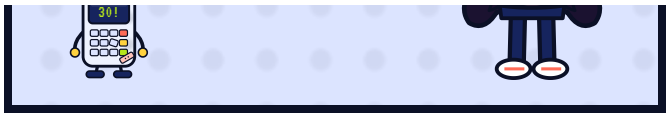
Sum of squares = 3.60

MSE = 3.60 ÷ 5 = 0.12





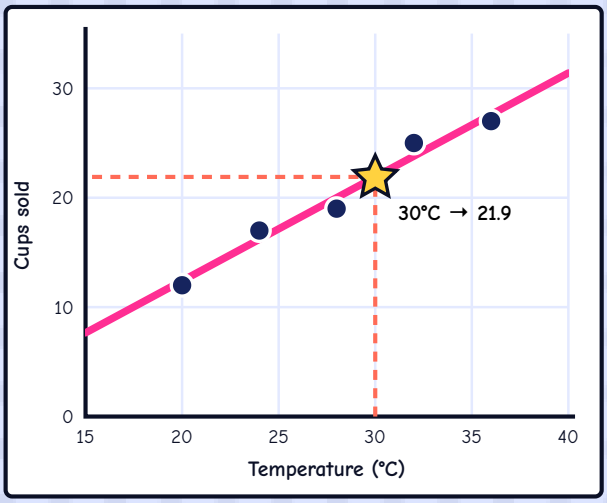




PREDICTION TIME

$$y = 0.95 \times 30 - 6.6$$
$$= 28.5 - 6.6$$
$$= 21.9$$
$$\approx 22 \text{ CUPS}$$

Using the line to guess a new day is called a **prediction**.



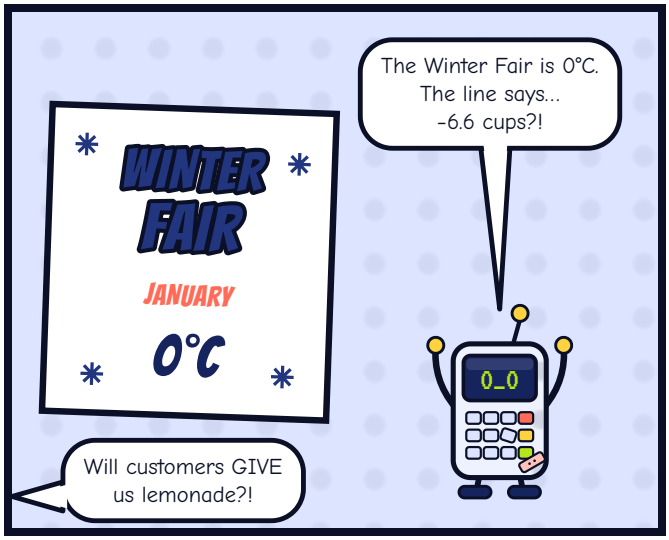
WHAT WE LEARNED

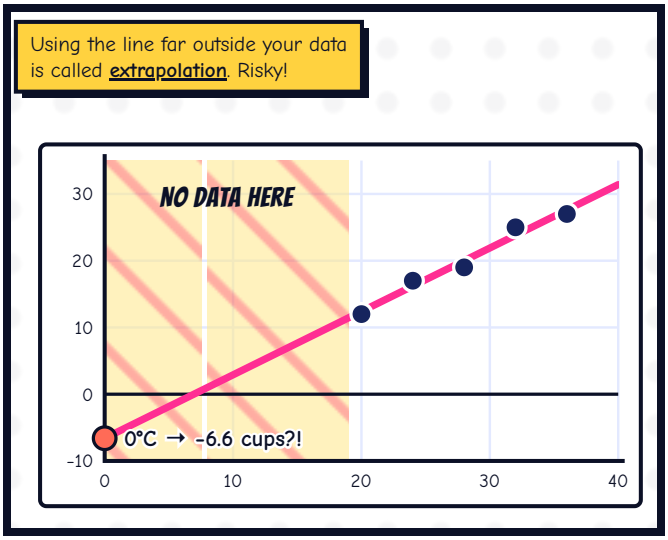
- Best line: $y = 0.95 \cdot x - 6.6$ (MSE 0.72).
- Slope 0.95: each extra °C ≈ one more cup.
- **Prediction**: plug a new x into the line.
- Saturday, 30°C → about 22 cups. It worked!

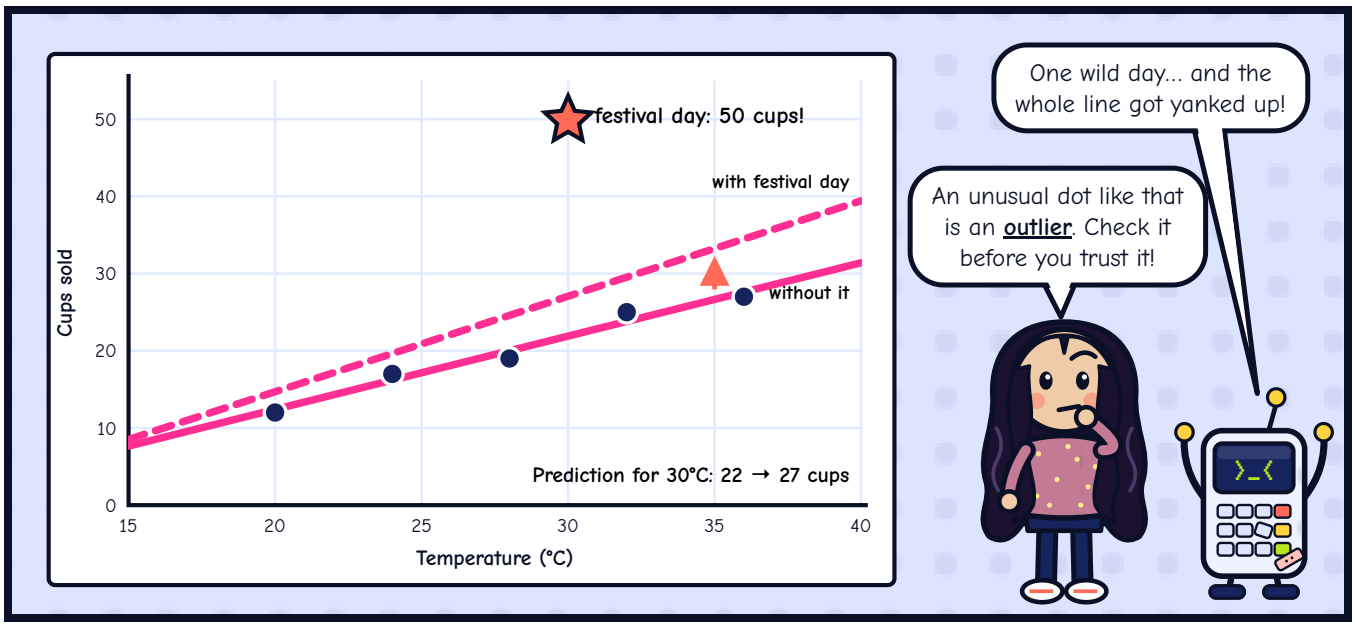
CHAPTER 6

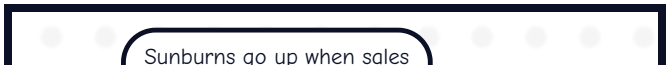
WHERE THE LINE LIES

In which the line gets one very silly answer.

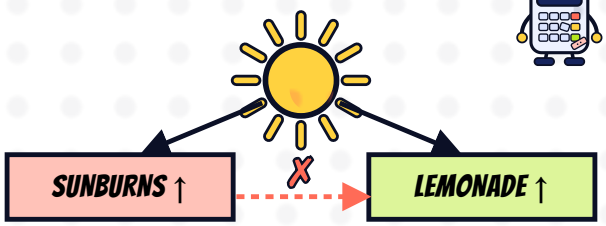






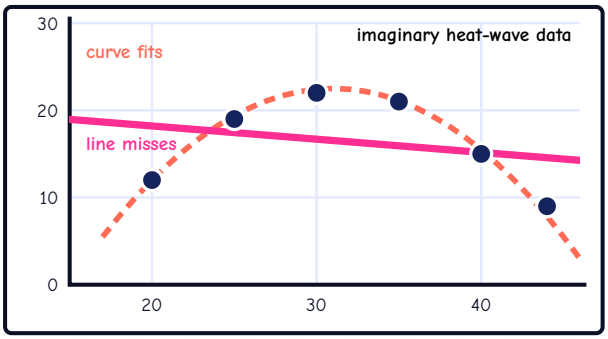


go up... so should I give customers sunburns?!



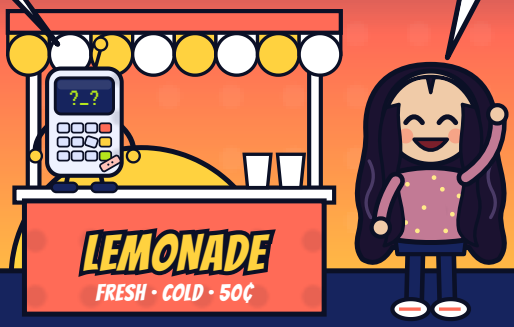
NO! The heat causes both. Moving together (**correlation**) isn't the same as causing.

inside, the dots BEND. A straight line can't bend. Other models can.



What if we ALSO used the day of the week? And the price?

More inputs in one model: **multiple linear regression**. Our next adventure!



TO BE CONTINUED...

WHAT WE LEARNED

- Trust the line only near your data (no wild **extrapolation**).
- Watch for **outliers**: one weird day can drag the whole line.
- Moving together (**correlation**) doesn't prove one causes the other.
- Curvy patterns need models that can bend.

THE END

OF THE BEGINNING. NEXT: THE CHEAT SHEET & QUIZ

LINO'S NOTEBOOK · TEAR-OUT PAGE

THE CHEAT SHEET

THE RECIPE

$$y = M \cdot x + B$$

x is the **input** (feature) you know. y is the **output** (target) you want to predict. Xiri's best line: y = 0.95·x − 6.6.

M · SLOPE

How steep the line is. How much y changes when x goes up by 1. Here: about one extra cup per extra °C.

B · INTERCEPT

Slides the whole line up or down. It is the line's value at x = 0.

RESIDUAL

Real value minus the line's guess. The leftover error for one data point.

MSE

Mean Squared Error: square every residual, then average. Lower is better. Guess: 8.2. Best line: 0.72.

GRADIENT DESCENT

Find the lowest MSE by stepping downhill again and again. The **learning rate** is the step size: too big overshoots, too small crawls.

LEAST SQUARES

For a straight line, a formula can jump straight to the lowest MSE. Big models without a formula use gradient descent instead.

PREDICTION

Plug a new x into the line. Saturday at 30°C: 0.95 × 30 − 6.6 = 21.9, so about 22 cups.

THREE CAUTIONS

- **Extrapolation:** the line only knows 20–36°C. At 0°C it says −6.6 cups.
- **Outliers:** one weird day can drag the whole line.
- **Correlation isn't causation:** moving together doesn't mean one causes the other.

NEXT UP

More inputs at once (temperature, day of the week, price) is **multiple linear regression**. Same idea, more knobs.

YOUR TURN

CAN YOU PREDICT?

Use Xiri's best line, **y = 0.95·x − 6.6**. Type your guess in cups, then check it or reveal the answer.

Q1. TOMORROW IS 26°C. HOW MANY CUPS SHOULD XIRI MAKE?

cups | CHECK | REVEAL ANSWER

Q2. TOMORROW IS 34°C. HOW MANY CUPS SHOULD XIRI MAKE?

cups | CHECK | REVEAL ANSWER

Q3. TOMORROW IS 22°C. HOW MANY CUPS SHOULD XIRI MAKE?

cups CHECK REVEAL ANSWER

BONUS. TOMORROW IS 45°C. HOW MANY CUPS SHOULD XIRI MAKE?

cups CHECK REVEAL ANSWER